Phase 4.25: Rank-34 TransitiveReduce Closure and Rank 1–40 Revisit

Tactix development report for Andrew

31 August 2026

1 Executive outcome

The remaining Rank-34 conflict-7,705 formula divergence has been localized and repaired. Pinned Kissat immediately learns a unit when TransitiveReduce finds a failed source literal, after retaining reductions already proved for that source. Tactix found the same failed walks but discarded the earlier tombstones and deferred the units to Probe. Tactix now matches the reference lifecycle and emits a native-LRAT BFS chain for each failed-source unit.

The repair closes the exact local state and materially improves the current Rank-34 baseline: 1,014.550 seconds falls to 804.709 seconds with a verified model. It does not recover the earlier 616.624-second trajectory or Kissat's 344.045 seconds. A uniform Rank 1–40 candidate solves all 40 instances, but totals 10,573.791 seconds versus 9,305.953 for the preceding homogeneous candidate. The general repair remains retained; the 13.6% aggregate regression is a downstream mechanism/trajectory problem, not grounds for a formula or rank gate.

2 Exact causal evidence

Both solvers entered the second conflict-7,705 TransitiveReduce pass with the same duplicate-sensitive 3,378-copy binary multiset. Tactix removed 41 copies and Kissat 39. The only excess Tactix deletions were `(-19396 v 19343)` and `(-19788 v 19343)`.

Selected-pair tracing showed the precise chronology. Kissat processes sources `-19788` and `-19396` before source `19343`. Each first walk reaches both polarities, so Kissat immediately fixes the source. Source `19343` then sees both destinations fixed and admits five candidates. Tactix deferred both units, admitted seven candidates for source `19343`, and deleted the two extra binaries.

Pinned `transitive.c` confirms that its failed branch compacts earlier reductions, calls `kissat_learned_unit`, and propagates before continuing. This is an unconditional rule lifecycle, not a scheduling heuristic.

3 Repair and correctness

The Tactix BFS now records the id of each live binary that first reaches a literal. Under native LRAT, that propagation order is the RUP hint chain for the new unit; the contradiction-closing edge is last. Earlier deletions are committed, the failed source is assigned, and the canonical or dosed surface reaches root before the next source. DefaultPolicy owns propagation and H publication; TransitiveReduce continues to own only its semantic and proof delta.

Causal job 1657559 makes all four begin/end binary snapshots exact, with counts 3,355, 3,312, 3,378, and 3,339. It emits units `-19788` and `-19396` in the second pass and finishes with the same 14,378 active variables and 14,259 heap entries as Kissat.

Correctness evidence after the final codebase cleanup is:

- Release 76/76 and Trace 75/75;
- four 50,000-formula TransitiveReduce brute-force modes in Release, Trace, and ASan/UBSan;
- an externally accepted native-LRAT failed-source proof;
- a focused fixture in which one earlier deletion must survive a later failed-source unit; and
- the affected ASan/UBSan, proof/strategy boundary, configuration, slot pose, and campaign-pinning gates.

No retained change adds a formula, rank, benchmark, identity, size, density, classification, admission, score, heap, conflict, or effort gate. Kissat creation order remains the unconditional production default; the CaDiCaL order is diagnostic only. BVE, Factor, forward subsumption, and Vivify remain at uniform 100/50/100/100 permille defaults. N1–N4 remain intact.

4 Rank-34 production result

Checked job 1657561 uses immutable Release SHA-256 `c7831464...e4066`; the campaign manifest records the full digest. It reports SAT in 804.709 seconds, 8,641,379 conflicts, 16,866,288 decisions, 1,793,431,878 propagations, and 51 transitive units. The independent model checker accepts all 120,226 clauses. Solver peak RSS is 133,756 KiB, and all certificate and derived artifacts are deleted.

Rank-34 run	Seconds	Ratio to current	Ratio to Kissat
Preceding mechanism-correct baseline	1,014.550	1.261x	2.949x
Transitive-unit repair	804.709	1.000x	2.339x
Earlier evaluated candidate	616.624	0.766x	1.792x
Pinned Kissat	344.045	0.428x	1.000x

The rule itself accounts for only 199 ms in the repaired run. The residual gap is search trajectory and inherited pre-Eliminate heap pose, not TransitiveReduce throughput.

5 Homogeneous Rank 1–40 validation

Job 1657564 runs one immutable production Tactix binary for all 40 ranks. No Kissat comparator was rerun; the fully checked pinned campaign remains the immutable authority. All 22 SAT and 18 UNSAT solver tasks return the expected verdict without timeout. All 22 SAT models are independently verified. At this report snapshot, 10 of 18 UNSAT proofs are verified and eight continue in the background; every completed result is valid and residue-free. Provisional UNSAT timing is intentionally not blocked on checker latency.

Measure	Candidate	Preceding	Ratio	Kissat	Ratio
Total seconds	10,573.791	9,305.953	1.136x	7,675.236	1.378x
Geometric mean	–	–	1.103x	–	1.366x
SAT seconds	5,034.213	4,083.055	1.233x	4,057.914	1.241x
UNSAT seconds	5,539.578	5,222.898	1.061x	3,617.322	1.531x
Solved	40/40	40/40	–	40/40	–

The largest regressions are Rank 17 (449.150 versus 168.543 seconds), Rank 19 (452.403 versus 163.790), Rank 21 (289.039 versus 131.230), Rank 27 (570.594 versus 373.282), Rank 34 (807.219 versus 616.624), and Rank 35 (602.741 versus 458.514). Rank 40 improves from 505.891 to 420.114 seconds. Rank 36 remains close to Kissat at 386.217 versus 368.510 seconds. Peak process-tree RSS is essentially unchanged at 16,987,860 KiB, still owned by Rank 10.

6 Codebase revisit

The retained diff was reread for stale scheduling state, proof/strategy boundary violations, diagnostic leakage, duplicated root lifecycle, and obsolete formula profiles. The revisit found and fixed one behavior-neutral duplication: dosed entries already consumed direct and propagated roots per candidate, but the outer rule boundary rescanned the same trail suffix. Batch entries retain their boundary publication. It also corrected a legacy diagnostic environment name in the proof fixture and added the exact prior-deletion-plus-failed-unit regression.

No residual `bve_min_bound`, `bve_track_min_bound`, or `transitive_failed` state remains. Unsupported Dose5, Factor5, NoBve, NoBveLight, and StartupFactorOnly profiles remain absent. The cleanup does not change formula state, ordering, scoring, effort, or admission.

The detailed report and full per-rank table are:

- `benchmarks/reports/2026-08-31-transitive-unit-rank1-40-codebase-revisit.md`;
- `benchmarks/runs/20260831-3630-tierd-ranks1-40-transitive-unit/comparison-live.md`; and
- `benchmarks/reports/2026-08-30-rank34-sat-divergence.md`.

7 Decision and next work

Retain the TransitiveReduce repair and every other causally proved general repair. Do not promote the current aggregate as a performance acceptance and do not restore a hidden CaDiCaL heuristic or formula-specific profile. The next convergence pass should localize the earliest common-state owner shared by the large R17/R19/R21 regressions, then return to Rank 34's inherited pre-Eliminate heap pose. Factor and formula classification remain downstream or exonerated.

Memory remains a parallel representation priority: Rank 10 is near 17 GiB and several medium instances remain two to three times Kissat's RSS. The current checkpoint nevertheless satisfies the requested implementation closure, homogeneous coverage run, codebase revisit, durable progress update, and major Andrew report.